

LUCAS TAN

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EDUCATION

National University of Singapore Bachelor of Engineering (Honours), Mechanical Engineering	Aug 2024 - Present
• Industry 4.0 Specialisation ◦ Feedback Control Systems; Cyber-Physical Systems; Automation in Manufacturing; Industrial Control Systems with AI • Minor in Applied AI (D&E) ◦ Machine Learning Applications & Artificial Intelligence Applications	

TECHNICAL SKILLS

- CAD: SOLIDWORKS, Fusion 360, Autodesk Inventor
- Programming & Computing: Python, C/C++ (Arduino), MATLAB
- Embedded Systems: Raspberry Pi, Arduino Uno
- Simulation: SOLIDWORKS Simulation, STAAD.Pro

WORK EXPERIENCE

Engineering Intern, R.J. Crocker Consultants Pte Ltd.	Apr 2024 - May 2024
• Modelled steel brackets and permanent blisters using STAAD.Pro to support the design of safe jacking points for an LRT bearing replacement project in Punggol • Accompanied engineers at North-South Corridor site, gaining practical exposure to technical assessments, construction processes, and engineering project workflows • Cross-referenced numerical values across physical records, digital documents, and information issued to external recipients to identify discrepancies and maintain consistency	
System Network Administrator, Singapore Armed Forces	Mar 2022 - Mar 2024
• Managed computer network infrastructure supporting the planning and execution of military operations • Supported the maintenance of virtual machines using Nutanix and Hyper-V Manager and administered Windows Server and Active Directory across 100+ mission-critical endpoints • Conducted system monitoring and incident response that maintained 99% system uptime	

ENGINEERING PROJECTS

Infineon Supply Chain Case Challenge Participant	Jun 2026 - Jul 2026
• Developed a graph-based framework to analyse semiconductor logistics, utilising kNN and GraphSAGE to map dynamic supply chain routes • Implemented a modified Dijkstra's algorithm to optimise routing, integrating a Multi-Layer Perceptron for route pruning and efficiency forecasting	
EG1311 Arduino Obstacle-Course Robot Project 1st in Class	Aug 2025 - Nov 2025
• Placed 1st in class as part of a four-member team that designed, fabricated and programmed a lightweight Arduino-controlled obstacle-course robot • Iteratively prototyped and tested wheel, chassis and launching-mechanism designs, developing laser-cut acrylic wheels with rubber rims and a lightweight chassis to improve traction, structural integrity and weight efficiency • Integrated and troubleshooted Arduino-controlled motors, servo mechanisms and electrical circuits, resolving loose connections and insufficient motor current to improve reliability and performance	

CE2155 Structural Mechanics & Materials Competition | 1st in Cohort **Jan 2025 - Apr 2025**

- Placed 1st in competition and received the HDB Prize for the design of a lightweight paper wind tower
- Collaborated with team members and applied structural mechanics principles, prototyping and load testing to optimise structural strength, stability and material efficiency

CE1103 Tower Crane Design and Analysis Project | 1st in Cohort **Aug 2024 - Nov 2024**

- Placed 1st in module by collaborating in a four-member team on the structural design, ETABS analysis, soil testing, physical construction, technical reporting and load testing of a basswood tower crane with a sand-clay foundation
- Contributed to modelling and analysing the structural scheme in ETABS, evaluating load paths, member forces and potential failure modes to support the crane's structural design and predicted load capacity

National Junior Robotics Competition (Sumo Bot) | 3rd in Nationals **Mar 2016 - Sep 2016**

- Attained 3rd place nationally by designing, building and programming a LEGO Mindstorms EV3 sumo robot within strict weight and dimensional constraints
- Researched and evaluated different gear ratios to optimise torque within motor limits, improving the robot's pushing power and competitive performance
- Refined and tested multiple designs, incorporating low-profile LEGO parts to prevent lifting and gain leverage beneath opponents

AWARDS & CERTIFICATES

- HDB Prize (CE2155 Structural Mechanics & Materials Competition)
- Dassault Systèmes SOLIDWORKS Certifications: CSWP (Mechanical Design), CSWA-E (Electrical), CSWA-S (Simulation), CSWA-AM (Additive Manufacturing)
- Cisco CCNA: Introduction to Networks Certification

CO-CURRICULAR EXPERIENCE

Externals Member, NUS Civil Engineering Club **Aug 2024 - May 2025**

- Spearheaded the end-to-end organisation of a site visit to BCA Academy, liaising directly with external stakeholders and designing promotional materials to recruit participants, coordinate logistics, gather post-visit feedback and registration information
- Coordinated participant registration and administrative logistics for industry site visits to Tuas Port and Elias MRT, managing sign-up communications and maintaining accurate records
- Consolidated and summarised post-visit survey responses, communicating participants' feedback to external partners to support programme evaluation and future site-visit planning

Green Audit Member, NUS SAVE **Aug 2024 - Dec 2024**

- Conducted sustainability audits for all 14 participating teams at NUS RAG 2024 by reviewing material measurements and supporting evidence to assess their use of recycled and reused materials against event sustainability guidelines
- Reviewed sustainability scoring methodology and recommended improvements for the following year's audit cycle

ADDITIONAL INFORMATION

- **Languages:** Proficient in English and Chinese (spoken and written)
- **Additional skills:** Microsoft Office (Excel, Word, PowerPoint)
- **Hobbies and Interests:** Running, Repairing old electronics